

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Examination, 2020

URP 2113
(Site and Area Planning)

Full Marks: 120

Time: 1.50 hrs

- N.B.**
- i) Answer any **two** questions from each section in separate scripts.
 - ii) Figures in the right margin indicate full marks.
 - iii) Assume data if necessary

Section – A

1. (a) Describe the four basic models of site planning with the help of an appropriate example. (10)
(b) Describe a site of your choice on the following criteria: (10)
(i) Spirit (ii) Character (iii) Nature of the project (iv) Behavioral studies
(c) Explain “SMART” growth in the context of site planning. (10)
- 2 (a) List out the contents of a Site Inventory Checklist. (15)
(b) “Structural and drainage characteristics are of utmost importance”- mention and elaborate on any five such characteristics. (15)
3. (a) What are the climatic elements in site planning and how can one modify their presence? (10)
(b) Prepare a wind rose diagram with assumed data and provide necessary measures that can be taken for proper wind flow. (10)
(c) Elaborate on the following: (10)
(i) Guiding principles for Recreation, Parks and Open Space (RPOS) planning
(ii) RPOS challenges

Section – B

4. (a) “Site planning is teamwork and involved with several professions viz. Urban Planning, Architecture, Landscape Architecture and Civil Engineering”- briefly explain with necessary example. (10)
(b) Why is the neighborhood addressed as a nucleus for the development of the local social life of the environment? (10)
(c) “The size of the neighbourhood unit is expressed as the size of the population and the extent of the geographical area”-explain with necessary example. (10)
5. (a) Why should site planning be carried out in connection with nature? (07)
(b) Write down any three principles of block design in site planning. (08)
(c) How are “density” and “setbacks” integrated as an element of residential subdivision in the site and area planning? (15)
6. (a) Write about the benefits of using technologies for site and area planning in the modern era with some limitations. (06)
(b) Annotate any three of the following pattern of subdivision: (24)
(i) Gridiron pattern (ii) Modified grid (iii) Curvilinear pattern
(iv) Rectilinear Pattern (v) Cul-de-sac pattern

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2ndYear 1st Term Examination, 2020
URP 2115
(Urban Planning Techniques)

Full Marks: 120

Time: 1.5 hrs

- N.B.** i) Answer any Two questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume data if necessary

Section – A

1. (a) Narrate the significance and scope of urban development plan in controlling unplanned growth. (15)
(b) “Urban planning decisions are based on certain planning principles, standards, norms and the decisions are implemented using certain planning tools”-explain using required example. (15)
2. (a) Master Plan is known as “Land use plan”-why? Elaborate the preparation and execution process of Master Plan executed by empowered authority in Bangladesh. (15)
(b) Make a comparison between subject plan and action plan – two forms of local plan with appropriate example. (08)
(c) Structure plan provides more flexible and open-ended measures-explain with example. (07)
3. (a) Critically analyze “Land Pooling” technique of residential area development with its applicability in Bangladesh. (10)
(b) “Land Banking technique is initiated to control urban growth and land speculation” –explain using necessary example. (10)
(c) Make a summery on “Dhaka Metropolitan Building Construction Rule, 2008” considering vital planning provisions suggested in the rule. (10)

Section – B

4. (a) How projections are mislabelled as forecasts? Describe the linear and logistic curves with their advantages and disadvantages. (15)
(b) Illustrate the concept of “Location Quotient Approach” with its conditions. What are the major problems associated with it? (15)
5. (a) Which factors are considered for collection and evaluation of housing data? Describe them. (10)
(b) Mention the steps of Mapping and Interpretation methods of Build Out analysis. (10)
(c) Define “FAR” with their symbolic meaning. Briefly describe the concept of Land Suitability Analysis. (10)
6. (a) What do you mean by Parks, Recreation Center and Open-Space (PROS)? Mention the importance of the need assessment of PROS. (10)
(b) According to your opinion, what would be the possible solution between urban renewal and demolition? Support with required example. (10)
(c) Diagrammatically represent Douglas’ Level of Intervention of urban renewal. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Examination, 2020

URP 2117
(Sociology for Planners)

Full Marks: 120

Time: 1.5 hrs

- N.B.** i) Answer any Two questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume data if necessary

Section – A

1. (a) What is meant by sociology? (10)
(b) Discuss the importance of studying sociology for the students of URP (20)
2. (a) What is called society and community? Mention the characteristics of society and community. (15)
(b) Distinguish between society and community with examples. (15)
3. (a) What is culture? Discuss the role of culture in our society. (10)
(b) Define Association and Institution. Mention the characteristics of Association and Institution with examples. (10)
(c) Explain the impacts of urbanization on our social life. (10)

Section – B

4. (a) What is crime and juvenile delinquency? Distinguish between crime and delinquency. (15)
(b) What do you mean by family? Discuss the functions of family in social life. (15)
5. (a) Explain the features of urban social structure of Bangladesh. (15)
(b) Describe the role of agencies responsible for controlling the criminal activities in Bangladesh. (15)
6. (a) What is marriage? Explain the importance of marriage in maintaining and keeping social order. (15)
(b) What are the social challenges for economic development of Bangladesh? Explain with appropriate examples. (15)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Examination, 2020
URP 2191
(Statistics for Planners - I)

Full Marks: 120

Time: 1.5 hrs

- N.B.** i) Figures in the right margin indicate full marks.
iii) Answer any **two** questions from each section in separate script.

Section – A

1.	(a)	Define statistics in your own words. Mention the use of statistics for Urban and Regional Planners.	(10)														
	(b)	Explain different levels of measurement of statistical data with appropriate examples.	(08)														
	(c)	Draw a histogram using the following data.	(12)														
		<table><tr><td>Class</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td></tr><tr><td>Frequency</td><td>5</td><td>10</td><td>15</td><td>17</td><td>9</td><td>3</td></tr></table>	Class	10-20	20-30	30-40	40-50	50-60	60-70	Frequency	5	10	15	17	9	3	
Class	10-20	20-30	30-40	40-50	50-60	60-70											
Frequency	5	10	15	17	9	3											
2.	(a)	What are the common measures of data centrality and data variability? Explain why both are important for exploring data.	(12)														
	(b)	Following table shows grouped data for heights of 100 randomly selected male students.	(18)														
		<table><tr><td>Height (inches)</td><td>Frequency</td></tr><tr><td>59.5-62.5</td><td>5</td></tr><tr><td>62.5-65.5</td><td>18</td></tr><tr><td>65.5-68.5</td><td>42</td></tr><tr><td>68.5-71.5</td><td>27</td></tr><tr><td>71.5-74.5</td><td>8</td></tr></table>	Height (inches)	Frequency	59.5-62.5	5	62.5-65.5	18	65.5-68.5	42	68.5-71.5	27	71.5-74.5	8			
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		Calculate skewness and comment on the height distribution.															
3.	(a)	Explain the major characteristics of a good classification with reference to number of classes, class intervals and class limits.	(12)														
	(b)	The following are monthly revenue in thousands of taka of 30 street vendors in Khulna. 45, 30, 19, 7, 33, 28, 15, 9, 11, 21, 16, 19, 24, 33, 35, 27, 18, 14, 22, 17, 26, 38, 17, 23, 16, 15, 19, 40 Represent the data in tabular form with range and calculate median monthly income.	(18)														

Section – B

4.	(a)	Write short notes on the followings with examples: (i) Descriptive statistics vs. Inferential statistics (ii) Limitations of Statistics (iii) Statistical reasoning vs. Probabilistic reasoning	(15)
	(b)	Two coins are tossed. Answer the following: (i) What is the sample space? (ii) Suppose, E is the event of tossing exactly two heads. F is event of getting at least 1 head. G is the event of tossing exactly two tails. What is P(E), P(F) and P(G)?	(05)
	(c)	A bag contains 4 cards numbered 2, 4, 6, 9. A second bag contains 3 cards	(10)

		numbered 2, 3, 6. One card is drawn at random from each bag. (i) Draw a tree diagram for the experiment. (ii) From the tree diagram, calculate the probability that the two numbers obtained: (A) have different values; (B) are both even; (C) are both prime; (D) have a sum greater than 5 and (E) have a product greater than 16.	
5.	(a)	Write down the characteristics and shortcomings of classical probability.	(10)
	(b)	Discuss Bayes' Theorem in terms of prior probability and posterior probability.	(10)
	(c)	Jar B_1 contains 2 white and 3 black balls and Jar B_2 contains 3 white and 4 black balls. One jar is selected at random and a ball is drawn from it. If the ball drawn is found black, find the probability that the jar chosen was B_1 .	(10)
6.	(a)	What are the characteristics of a Normal probability distribution? Explain with necessary illustrations.	(10)
	(b)	During the rush hours cars arrive at a city centre car parks at the rate of 35 per 60 minutes. (i) What is the probability of exactly 5 arrivals over a 10 minutes time period. (ii) What is the probability of exactly one arrival over a 10 minutes time period.	(13)
	(c)	What is Poisson probability distribution? Mention the assumptions of Poisson probability distribution.	(07)

Khulna University of Engineering & Technology
 Department of Urban and Regional Planning
 BURP 2nd Year 1st Term Regular Examination, 2020
CE 2107
 (Construction Technology and Materials)

Full Marks: 120

Time: 1.5hrs

- N.B.**
- i) Figures in the right margin indicate full marks.
 - ii) Assume suitable data if necessary.
 - iii) Answer any two questions from each section in separate script.

Section – A

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|---|--|----|
| 1 | (a) What is Cement? Why cement is used widely instead of lime in Civil Engineering Projects? | 08 |
| | (b) Classify Cement according to EN 197-1 and ASTM-C150. Explain these types of cements shortly. | 12 |
| | (c) What is meant by false setting? What are the causes of false setting? What are the remedies of this problem? | 10 |
| 2 | (a) Define Sand. Write down the classification of sand according to size of sand particle. Which sand does an engineer propose for making cement mortar? | 09 |
| | (b) What is “Fineness Modulus” of Sand? Describe the effect of bulking of sand. | 06 |
| | (c) There are two samples of sand in a construction project. One sand is from Kalikair (FM= 2.90) and another sand is from Sunamganj (FM= 2.15). These two sands are mixed to produce a sample of 1235gm. If the combined fineness modulus is 2.49, Calculate the amount of each variety of the sand in the mix. | 10 |
| | (d) How does one determine the moisture content in sand in the field? | 05 |
| 3 | (a) Define concrete. Why concrete is the most widely used Engineering Material? | 10 |
| | (b) What is meant by the term “Curing of Concrete”? What are the effects of Water/Cement ratio on the strength of concrete? | 10 |
| | (c) What are the field test to identify the good bricks? Describe the class of bricks. | 10 |

Section – B

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|---|---|----|
| 4 | (a) Define “Foundation”. What are the purposes of foundation? | 08 |
| | (b) When do you suggest trapezoidal shape of combined footing in place of rectangular shape? | 08 |
| | (c) What is meant by “Safe bearing Capacity” of a soil? Describe plate load test with necessary illustrations for determining the safe bearing capacity of soil at site. | 14 |
| 5 | (a) What is load bearing and non-load bearing wall? Why bonds are important for masonry walls? | 08 |
| | (b) What is efflorescence? How can it be removed from the surface of the wall? | 07 |
| | (c) List the governing factors for the selection of suitable type of floor. | 07 |
| | (d) What is cavity wall? What are the main advantages of cavity wall? | 08 |
| 6 | (a) What are the facilities of segmental construction techniques in bridge construction? | 08 |
| | (b) What are the objectives of plastering. Describe the defects found in plaster. Briefly explain the different types of plasters adopted for internal and external finishing of wall surfaces. | 12 |
| | (c) What are the components of plumbing? Draw the schematic diagram of Domestic incoming Electrical service. | 10 |
